

Task-4

MapReduce Programming

Initial Steps

1. Create the EMR cluster.
2. Install mr job as root user: **pip install mrjob**
3. Make directory 'yellow' as Hadoop user: **mkdir yellow**
4. Download the all the csv files in the folder yellow. I downloaded and ran on 1 file as the job got killed due to too long for 4 csv files and also all 6 files were too large for the Custer to store.

```
[hadoop@ip-172-30-0-134 yellow]$ wget https://nyc-tlc-upgrad.s3.amazonaws.com/yellow_tripdata_2017-01.csv
--2023-10-09 17:32:17-- https://nyc-tlc-upgrad.s3.amazonaws.com/yellow_tripdata_2017-01.csv
Resolving nyc-tlc-upgrad.s3.amazonaws.com (nyc-tlc-upgrad.s3.amazonaws.com)... 52.217.142.137, 52.217.142.65, 3.5.8.111, ...
Connecting to nyc-tlc-upgrad.s3.amazonaws.com (nyc-tlc-upgrad.s3.amazonaws.com)|52.217.142.137|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 914029548 (872M) [text/csv]
Saving to: 'yellow_tripdata_2017-01.csv'

100K[=====] 914,029,548 25.6MB/s in 48s

2023-10-09 17:32:57 (21.6 MB/s) - 'yellow_tripdata_2017-01.csv' saved [914029548/914029548]

[hadoop@ip-172-30-0-134 yellow]$ wget https://nyc-tlc-upgrad.s3.amazonaws.com/yellow_tripdata_2017-02.csv
--2023-10-09 17:33:00-- https://nyc-tlc-upgrad.s3.amazonaws.com/yellow_tripdata_2017-02.csv
Resolving nyc-tlc-upgrad.s3.amazonaws.com (nyc-tlc-upgrad.s3.amazonaws.com)... 3.5.28.119, 16.182.48.281, 52.216.211.217, ...
Connecting to nyc-tlc-upgrad.s3.amazonaws.com (nyc-tlc-upgrad.s3.amazonaws.com)|3.5.28.119|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 863487858 (823M) [text/csv]
Saving to: 'yellow_tripdata_2017-02.csv'

100K[=====] 863,487,858 26.6MB/s in 37s

2023-10-09 17:33:37 (22.2 MB/s) - 'yellow_tripdata_2017-02.csv' saved [863487858/863487858]

[hadoop@ip-172-30-0-134 yellow]$ aws s3 cp s3://jayazams3bucket/mrtask_c.py /home/hadoop/pythonfiles
fatal error: An error occurred (403) when calling the HeadObject operation: Forbidden
[hadoop@ip-172-30-0-134 yellow]$ wget https://nyc-tlc-upgrad.s3.amazonaws.com/yellow_tripdata_2017-03.csv
--2023-10-09 17:33:42-- https://nyc-tlc-upgrad.s3.amazonaws.com/yellow_tripdata_2017-03.csv
Resolving nyc-tlc-upgrad.s3.amazonaws.com (nyc-tlc-upgrad.s3.amazonaws.com)... 52.217.120.249, 52.217.121.141, 52.217.126.137, ...
Connecting to nyc-tlc-upgrad.s3.amazonaws.com (nyc-tlc-upgrad.s3.amazonaws.com)|52.217.120.249|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 969899825 (929M) [text/csv]
Saving to: 'yellow_tripdata_2017-03.csv'

100K[=====] 969,899,825 23.2MB/s in 39s

2023-10-09 17:34:31 (24.0 MB/s) - 'yellow_tripdata_2017-03.csv' saved [969899825/969899825]

[hadoop@ip-172-30-0-134 yellow]$ wget https://nyc-tlc-upgrad.s3.amazonaws.com/yellow_tripdata_2017-04.csv
--2023-10-09 17:34:34-- https://nyc-tlc-upgrad.s3.amazonaws.com/yellow_tripdata_2017-04.csv
Resolving nyc-tlc-upgrad.s3.amazonaws.com (nyc-tlc-upgrad.s3.amazonaws.com)... 52.217.121.34, 52.217.121.141, 52.217.126.137, ...
Connecting to nyc-tlc-upgrad.s3.amazonaws.com (nyc-tlc-upgrad.s3.amazonaws.com)|52.217.121.34|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 946349441 (903M) [text/csv]
Saving to: 'yellow_tripdata_2017-04.csv'

100K[=====] 946,349,441 22.1MB/s in 38s

2023-10-09 17:35:12 (23.7 MB/s) - 'yellow_tripdata_2017-04.csv' saved [946349441/946349441]

[hadoop@ip-172-30-0-134 yellow]$ wget https://nyc-tlc-upgrad.s3.amazonaws.com/yellow_tripdata_2017-05.csv
--2023-10-09 17:35:14-- https://nyc-tlc-upgrad.s3.amazonaws.com/yellow_tripdata_2017-05.csv
Resolving nyc-tlc-upgrad.s3.amazonaws.com (nyc-tlc-upgrad.s3.amazonaws.com)... 52.216.245.284, 52.217.123.137, 52.217.286.49, ...
Connecting to nyc-tlc-upgrad.s3.amazonaws.com (nyc-tlc-upgrad.s3.amazonaws.com)|52.216.245.284|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 923945526 (889M) [text/csv]
Saving to: 'yellow_tripdata_2017-05.csv'

58M[=====] 558,673,837 20.1MB/s in 24s

Cannot write to 'yellow_tripdata_2017-05.csv' (Success).
[hadoop@ip-172-30-0-134 yellow]$ ls
yellow_tripdata_2017-01.csv yellow_tripdata_2017-03.csv yellow_tripdata_2017-05.csv
yellow_tripdata_2017-02.csv yellow_tripdata_2017-04.csv
[hadoop@ip-172-30-0-134 yellow]$ mv yellow_tripdata_2017-05.csv
[hadoop@ip-172-30-0-134 yellow]$ ls
yellow_tripdata_2017-01.csv yellow_tripdata_2017-03.csv
yellow_tripdata_2017-02.csv yellow_tripdata_2017-04.csv
[hadoop@ip-172-30-0-134 yellow]$ cd ..
[hadoop@ip-172-30-0-134 ~]$ pip install mrjob
Collecting mrjob
  Downloading https://files.pythonhosted.org/packages/8e/58/fc28ab743aba16e90736ad4e29694bd2ada7b879376ff149306d50c4e90/mrjob-0.7.4-py2.py3-none-any.whl (439kB)
    100K |#####| 440kB 3.1MB/s
Requirement already satisfied: PyYAML>=3.10 in /usr/local/lib64/python3.7/site-packages (from mrjob)
Installing collected packages: mrjob
Successfully installed mrjob-0.7.4
[hadoop@ip-172-30-0-134 ~]$ sudo -i

[hadoop@ip-172-30-0-134 ~]$ sudo -i

EEEEEEEEEEEEEEEEEEEE MMMMMMMMMM MMMMMMMMM RRRRRRRRRRRRRRR
E:EEEEEEEEEEEEEEEEEE M:EEEEEEEE M:EEEEEEEE R:EEEEEEEEEEEEEE
EE:EEEEEEEEEEEEEEEEEE M:EEEEEEEE M:EEEEEEEE R:EEEEEEEEEEEEEE
E:EEEEEEEEEEEEEEEEEE M:EEEEEEEE M:EEEEEEEE RR:EEEEEEEE R:EEEE
E:EEEEEEEEEEEEEEEEEE M:EEEEEEEE M:EEEEEEEE M:EEEEEEEE R:EEEE
E:EEEEEEEEEEEEEEEEEE M:EEEEEEEE M:EEEEEEEE M:EEEEEEEE R:EEEE
E:EEEEEEEEEEEEEEEEEE M:EEEEEEEE M:EEEEEEEE M:EEEEEEEE R:EEEE
E:EEEEEEEEEEEEEEEEEE M:EEEEEEEE M:EEEEEEEE M:EEEEEEEE R:EEEE
EE:EEEEEEEEEEEEEEEEEE M:EEEEEEEE M:EEEEEEEE M:EEEEEEEE R:EEEE
E:EEEEEEEEEEEEEEEEEE M:EEEEEEEE M:EEEEEEEE M:EEEEEEEE R:EEEE
EEEEEEEEEEEEEEEEEEEE MMMMMMMMMM MMMMMMMMM RRRRRRRRRRRRRRR

[root@ip-172-30-0-134 ~]# pip install mrjob
WARNING: Running pip install with root privileges is generally not a good idea. Try 'pip3 install --user' instead.
Collecting mrjob
  Downloading https://files.pythonhosted.org/packages/8e/58/fc28ab743aba16e90736ad4e29694bd2ada7b879376ff149306d50c4e90/mrjob-0.7.4-py2.py3-none-any.whl (439kB)
    100K |#####| 440kB 3.1MB/s
Requirement already satisfied: PyYAML>=3.10 in /usr/local/lib64/python3.7/site-packages (from mrjob)
Installing collected packages: mrjob
Successfully installed mrjob-0.7.4
[root@ip-172-30-0-134 ~]# exit
logout
[hadoop@ip-172-30-0-134 ~]$ vi mrtask_a.py
[hadoop@ip-172-30-0-134 ~]$ python3 mrtask_a.py /home/hadoop/yellow/
No configs found; falling back on auto-configuration
No configs specified for inline runner
Creating temp directory /tmp/mrtask_a.hadoop.20231009.173819.730239
Running step 1 of 2...
Killed
[hadoop@ip-172-30-0-134 ~]$ cd yellow/
[hadoop@ip-172-30-0-134 yellow]$ ls
yellow_tripdata_2017-01.csv yellow_tripdata_2017-03.csv
yellow_tripdata_2017-02.csv yellow_tripdata_2017-04.csv
[hadoop@ip-172-30-0-134 yellow]$ rm yellow_tripdata_2017-04.csv
[hadoop@ip-172-30-0-134 yellow]$ rm yellow_tripdata_2017-03.csv
[hadoop@ip-172-30-0-134 yellow]$ rm yellow_tripdata_2017-02.csv
[hadoop@ip-172-30-0-134 yellow]$ la
-bash: la: command not found
[hadoop@ip-172-30-0-134 yellow]$ ls
yellow_tripdata_2017-01.csv
```

Task a.

To find the vendor with most trips and the total revenue generated by them.

Solution:

When we run the mrtask_a.py using the command `python3 mrtask_a.py /home/Hadoop/yellow` the output is

```
[hadoop@ip-172-30-0-134 yellow]$ ls
yellow_tripdata_2017-01.csv
[hadoop@ip-172-30-0-134 yellow]$ cd ..
[hadoop@ip-172-30-0-134 ~]$ python3 mrtask_a.py /home/hadoop/yellow/
No configs found; falling back on auto-configuration
No configs specified for inline runner
Creating temp directory /tmp/mrtask_a.hadoop.20231009.175049.769112
Running step 1 of 2...
Running step 2 of 2...
job output is in /tmp/mrtask_a.hadoop.20231009.175049.769112/output
Streaming final output from /tmp/mrtask_a.hadoop.20231009.175049.769112/output...
2      82859254.39585336
Removing temp directory /tmp/mrtask_a.hadoop.20231009.175049.769112...
```

Vendor 2 has the highest trips and revenue generated is 82859254.39585336.

Task b.

To find the pick-up location which generated the most revenue.

Solution

Run the command `python3 mrtask_b /home/Hadoop/yellow`; output:

```
[[hadoop@ip-172-30-0-134 ~]$ vi b.py
[[hadoop@ip-172-30-0-134 ~]$ vi b.py /home/hadoop/yellow/
2 files to edit
[[hadoop@ip-172-30-0-134 ~]$ python3 b.py /home/hadoop/yellow/
No configs found; falling back on auto-configuration
No configs specified for inline runner
Creating temp directory /tmp/b.hadoop.20231009.182917.619785
Running step 1 of 2...
Running step 2 of 2...
job output is in /tmp/b.hadoop.20231009.182917.619785/output
Streaming final output from /tmp/b.hadoop.20231009.182917.619785/output...
132      12660835.9
Removing temp directory /tmp/b.hadoop.20231009.182917.619785...
[[hadoop@ip-172-30-0-134 ~]$
```

Location id `132` generated the max revenue of `12660835.9`.

Task c.

To calculate the count of each payment type

Solution:

Run the mrtask_c.py

Output:

```
[[hadoop@ip-172-30-0-134 ~]$ vi mrtask_c.py
[[hadoop@ip-172-30-0-134 ~]$ python3 mrtask_c.py /home/hadoop/yellow/
No configs found; falling back on auto-configuration
No configs specified for inline runner
Creating temp directory /tmp/mrtask_c.hadoop.20231009.185003.067499
Running step 1 of 2...
Running step 2 of 2...
job output is in /tmp/mrtask_c.hadoop.20231009.185003.067499/output
Streaming final output from /tmp/mrtask_c.hadoop.20231009.185003.067499/output...
"1"      6506189
"2"      3144926
"3"       46257
"4"      13447
"5"         1
Removing temp directory /tmp/mrtask_c.hadoop.20231009.185003.067499...
```

Task d.

To calculate average trip time for different pickup locations

Solution:

Running the mrtask_d.py

```
[hadoop@ip-172-30-0-10 yellow]$ cd ..
[hadoop@ip-172-30-0-10 ~]$ vi d.py
[hadoop@ip-172-30-0-10 ~]$ python3 d.py /home/hadoop/yellow/
No configs found; falling back on auto-configuration
No configs specified for inline runner
Creating temp directory /tmp/d.hadoop.20231010.082918.177548
Running step 1 of 1...
Job output is in /tmp/d.hadoop.20231010.082918.177548/output
Streaming final output from /tmp/d.hadoop.20231010.082918.177548/output...
1      10.3
10     39.29
100    14.19
101     9.76
102    13.43
105    19.41
106    15.45
107    13.27
108    12.65
109     3.79
11     17.7
110     3.18
111    11.24
112    14.48
113    13.79
114    14.79
115    14.63
116    14.98
117    33.36
118     5.04
119    15.01
12     20.97
120    14.53
121    16.02
122    28.11
123    13.05
124    20.72
125    14.28
126    25.73
127    14.98
128    14.84
129    14.08
13     17.91
130    29.11
131    17.37
132    38.53
133    15.3
134    14.38
135    11.63
136    10.94
137    13.35
138    32.29
139    14.46
14     14.22
140    13.38
141    11.71
142    12.89
143    12.46
144    15.6
145    14.65
146    14.69
147    14.91
148    15.74

42     12.69
43     14.38
45     17.64
46     9.09
47     12.21
48     13.79
49     13.34
5     0.37
50     13.35
51     14.39
52     16.19
53     14.34
54     14.61
55     16.87
56     15.89
57     10.31
58     5.28
59     16.98
6     5.71
60     12.04
61     13.09
62     13.74
63     14.19
64     10.7
65     15.47
66     16.91
67     36.45
68     14.05
69     13.86
7     13.25
70     22.0
71     20.42
72     15.05
73     8.09
74     12.19
75     12.03
76     30.24
77     18.08
78     12.04
79     14.40
8     15.73
80     13.43
81     15.58
82     14.16
83     15.8
84     0.32
85     19.73
86     16.18
87     18.43
88     19.41
89     12.89
9     37.35
90     12.97
91     21.23
92     13.51
93     24.18
94     12.03
95     16.25
96     20.0
97     14.95
98     13.59
Removing temp directory /tmp/d.hadoop.20231010.082918.177548...
[hadoop@ip-172-30-0-10 ~]$
```

Task e.

To Calculate the average tips to revenue ratio of the drivers for different pickup locations in sorted format.

Solution:

Running the mrtask_e.py

```
[hadoop@ip-172-30-0-10 ~]$ ls
d.py  yellow
[hadoop@ip-172-30-0-10 ~]$ vi e.py
[hadoop@ip-172-30-0-10 ~]$ python3 e.py /home/hadoop/yellow/
No configs found; falling back on auto-configuration
No configs specified for inline runner
Creating temp directory /tmp/e.hadoop.20231010.091329.546558
Running step 1 of 2...
Running step 2 of 2...
Job output is in /tmp/e.hadoop.20231010.091329.546558/output
Streaming final output from /tmp/e.hadoop.20231010.091329.546558/output...
176 0.2871
122 0.2441
30 0.2307
64 0.2126
117 0.2073
118 0.2041
245 0.2002
155 0.1965
191 0.1841
15 0.184
58 0.1614
139 0.1571
172 0.1524
111 0.1494
150 0.1449
252 0.1422
16 0.139
35 0.1374
210 0.1368
138 0.1339
175 0.1317
76 0.1285
52 0.1272
13 0.1254
125 0.1253
87 0.1245
194 0.1239
234 0.1222
9 0.1216
246 0.1214
54 0.1213
33 0.1209
249 0.1205
142 0.1197
131 0.1196
180 0.1194
40 0.1194
158 0.1193
187 0.1191
231 0.1188
90 0.1187
66 0.1187
113 0.1186
114 0.1179
79 0.1179
88 0.1174
170 0.1173
255 0.1169
233 0.1166
215 0.1165
161 0.1164
224 0.1163

285 0.0697
31 0.0697
240 0.0679
235 0.0678
149 0.0677
217 0.0654
82 0.0654
81 0.0651
258 0.0644
46 0.0641
218 0.0639
22 0.0634
156 0.0631
193 0.0625
92 0.0594
129 0.0589
39 0.0587
119 0.0572
72 0.0558
185 0.0557
165 0.0545
85 0.0545
212 0.0544
241 0.0541
83 0.054
168 0.0529
23 0.0524
183 0.0523
2 0.0515
169 0.0514
126 0.0512
136 0.0508
60 0.0505
173 0.0498
240 0.0492
242 0.0489
248 0.0485
174 0.0476
159 0.0466
18 0.0434
47 0.0427
6 0.0427
167 0.0412
59 0.0405
232 0.0389
206 0.038
69 0.0379
201 0.0373
153 0.0358
182 0.0292
38 0.027
86 0.0264
32 0.0212
254 0.0207
94 0.0192
251 0.0
187 0.0
110 0.0
189 0.0
105 0.0
5 0.0
Removing temp directory /tmp/e.hadoop.20231010.091329.546558...
[hadoop@ip-172-30-0-10 ~]$
```

Task f.

To show how revenue vary over time.

Solution:

Running the mrtask_f.py

Output.

Takes very long to execute and disconnects, happened many times.

```
[root@ip-172-30-0-169 ~]# ls
yellow
[root@ip-172-30-0-169 ~]# vi e.py
[root@ip-172-30-0-169 ~]# python3 e.py /root/yellow/
No configs found; falling back on auto-configuration
No configs specified for inline runner
Creating temp directory /tmp/e.root.20231010.105732.349109
Running step 1 of 2...
client_loop: send disconnect: Broken pipe
Aiethams-MacBook-Air:Upgrad aiethamvivek$ █
```